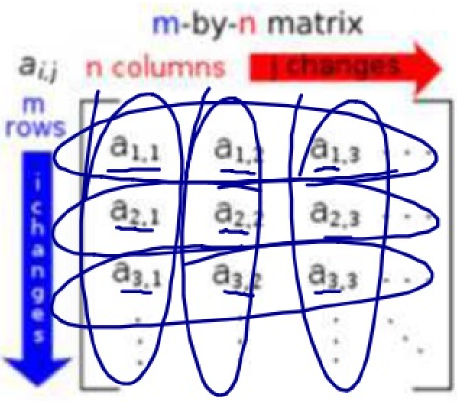
AP Calculus II
Notes P8.1
Gauss Elimination

Definition of a Matrix: A matrix is a rectangular array of number, variables, etc…with the individual terms known as **entries**. Matrices are described by the number of rows m and the number of columns n.



3 rows x 3 columns
3x3

Matrices are incredibly applicable in computer graphics, programming, cryptography, retail shopping, data storage/sorting and economics. One other use of matrices is solving systems of equations.

Elementary Row Operations

Ex. 1: Perform the following manipulations to the given matrix: $\begin{bmatrix} 2 & 5 & 3 & 6 \\ 1 & -2 & 0 & 8 \\ -3 & 4 & -1 & 3 \end{bmatrix}$.

a) Interchange R_1 and R_2

$$\begin{bmatrix} 1 & -2 & 0 & 8 \\ 2 & 5 & 3 & 6 \\ -3 & 4 & -1 & 3 \end{bmatrix}$$

b) Add -2 times R_2 to R_1

$$\begin{bmatrix} 0 & 9 & 3 & -10 \\ 1 & -2 & 0 & 8 \\ -3 & 4 & -1 & 3 \end{bmatrix}$$

Gauss Elimination

When it came to solving a system of equations, we used a technique known as Elimination/Linear Combination. With matrices, we will create an **augmented matrix**, which strips away all the variables and deals solely with the coefficients.

Ex. 2: Solve: $\begin{matrix} x+2y=9 \\ 2x+6y=22 \end{matrix}$ $\cdot -2$

$$\begin{array}{r} -2x-4y=-18 \\ +\ 2x+6y=22 \\ \hline 0x+2y=4 \end{array}$$

$0x+1y=2$

$x+4=9$

$x=5$

$\boxed{(5,2)}$

$$\left[\begin{array}{cc|c} 1 & 2 & 9 \\ 2 & 6 & 22 \end{array}\right]$$

$$-2\cdot R_1+R_2 \left[\begin{array}{cc|c} 1 & 2 & 9 \\ 0 & 2 & 4 \end{array}\right]$$

$$\tfrac{1}{2}\cdot R_2 \left[\begin{array}{cc|c} 1 & 2 & 9 \\ 0 & 1 & 2 \end{array}\right]$$

$y=2$

$x+4=9$

$x=5$

In this procedure, our goal will be to get the left side of the matrix to be in **row-echelon form**, which means that the first leading coefficient (nonzero) term in each row is always strictly to the right of the leading coefficient of the row above it. A better form is the **reduced row-echelon form** in which every leading coefficient is a 1 and is the only nonzero term of the column, which we will get to later.

$$\left[\begin{array}{ccc|c} 1 & 3 & 5 & -1 \\ 0 & -2 & 1 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right] \qquad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

Gauss Elimination with Back Substitution

Here are the steps to solving a system of equations using a Gauss Elimination:

1) Write the augmented matrix for the system of equations.
2) Interchange or multiply by a constant to get a 1 in the first entry in the first column.
3) Multiply the rows by a nonzero constant with the intent to produce zeros below the first entry in the first column. This constant is multiplied to the first row and then added to the other rows.
4) Repeat Step 3 to produce zeros below the second entry in the second column, third entry in the third column, and so on.
5) Once the matrix is in row-echelon form, the last row will have one variable solved. This can then be substituted back into the row above to solve for the next variable and so on to get the complete solution.

Ex. 3: Solve the following system of equation.

a) want 1 in top left corner

$$x - 2y + z = 0$$
$$2x + y - 3z = 5$$
$$4x - 7y + z = -1$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 2 & 1 & -3 & 5 \\ 4 & -7 & 1 & -1 \end{array}\right]$$

b) want 0's underneath the 1

what do I multiply the 1 by that when added to 2 will give 0?? $\boxed{-2}$

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 5 & -5 & 5 \\ 0 & 1 & -3 & -1 \end{array}\right] \begin{array}{l} \\ -2 \cdot R_1 + R_2 \\ -4 \cdot R_1 + R_3 \end{array}$$

c) Go down the diagonal and get a 1 there.

$\frac{1}{5} \cdot R_2$ (to get 1)

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 1 & -3 & -1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & -2 & -2 \end{array}\right] \begin{array}{l} \\ \\ -1 \cdot R_2 + R_3 \end{array}$$

Go down diagonal + get 1

$R_3 \div -2$

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$\boxed{z = 1}$

$y - 1 = 1$

$\boxed{y = 2}$

$x - 2(2) + 1 = 0$

$x - 3 = 0$

$\boxed{x = 3}$

$\boxed{(3, 2, 1)}$

1) Need 1 → 2) Need 0's underneath → 3) Go down the diagonal

Ex. 4: Solve the following system of equation.

a)
$$2x + y + z + w = 3$$
$$x + 2y + 3z + w = 5$$
$$x + 2y + z = 4$$
$$y + z + 2w = 0$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 5 \\ 2 & 1 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 4 \\ 0 & 1 & 1 & 2 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 5 \\ 0 & -3 & -5 & -1 & -7 \\ 0 & 0 & -2 & -1 & -1 \\ 0 & 1 & 1 & 2 & 0 \end{array}\right] \begin{array}{l} \\ -2R_1+R_2 \\ -1R_1+R_3 \\ \\ \end{array}$$

Flip R_2 + R_4

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 5 \\ 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & -2 & -1 & -1 \\ 0 & -3 & -5 & -1 & -7 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 5 \\ 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & -2 & -1 & -1 \\ 0 & 0 & -2 & 5 & -7 \end{array}\right]$$

$3 \cdot R_2 + R_4$

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 5 \\ 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1/2 & 1/2 \\ 0 & 0 & -2 & 5 & -7 \end{array}\right] \xrightarrow{2 \cdot R_3 + R_4} \left[\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 5 \\ 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1/2 & 1/2 \\ 0 & 0 & 0 & 6 & -6 \end{array}\right]$$

$x + 2 + 3 - 1 = 5, \; x = 1$

$y + 1 - 2 = 0, \; y = 1$

$z - \frac{1}{2} = \frac{1}{2}, \; z = 1$

$w = -1$

$$\boxed{(1, 1, 1, -1)}$$

b)
$$2a - 2b = -6$$
$$a - b + c = 1$$
$$3b - 2c = -5$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 2 & -2 & 0 & -6 \\ 0 & 3 & -2 & -5 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 0 & -2 & -8 \\ 0 & 3 & -2 & -5 \end{array}\right] -2R_1+R_2 \to \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & -2/3 & -5/3 \\ 0 & 0 & -2 & -8 \end{array}\right]$$

$$\to \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & -2/3 & -5/3 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

$c = 4$

$b - 8/3 = -5/3$

$b = 1$

$a - 1 + 4 = 1$

$a + 3 = 1$

$a = -2$

$$\boxed{(-2, 1, 4)}$$

AP Calculus II
Notes P8.1
Gauss-Jordan Elimination

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In this procedure, the elimination process is continued until the augmented matrix is in the reduced row-echelon form.

Ex. 1: Solve the following system of equations.

a) $\begin{aligned} x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{aligned}$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & -1 & -1 & -1 \end{array}\right] \begin{array}{l} \\ 1\cdot R_1 + R_2 \\ -2\cdot R_1 + R_3 \end{array}$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & \boxed{1} & 3 & 5 \\ 0 & 0 & 2 & 4 \end{array}\right] \begin{array}{l} \\ \\ 1\cdot R_2 + R_3 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

go back up

$$\begin{array}{l} -3\cdot R_3 + R_1 \\ -3\cdot R_3 + R_2 \\ \\ \end{array} \left[\begin{array}{ccc|c} 1 & -2 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right] \begin{array}{l} 2\cdot R_2 + R_1 \\ \\ \\ \end{array}$$

$x = 1$
$y = -1$
$z = 2$

b) $\begin{aligned} a - b + 2c &= 4 \\ a + c &= 6 \\ 2a - 3b + 5c &= 4 \\ 3a + 2b - c &= 1 \end{aligned}$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 1 & 0 & 1 & 6 \\ 2 & -3 & 5 & 4 \\ 3 & 2 & -1 & 1 \end{array}\right] \rightarrow \begin{array}{l} \\ -1\cdot R_1 + R_2 \\ -2\cdot R_1 + R_3 \\ -3\cdot R_1 + R_4 \end{array} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & -1 & 1 & -4 \\ 0 & 5 & -7 & -11 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & -2 \end{array}\right] \quad R_2 + R_3$$

$0 = -2$

never

$\therefore$ No Solution

Ex. 2: Solve the following system of equations:

$$2x+4y-2z=0$$
$$3x+5y=1$$

$$= \left[\begin{array}{ccc|c} 2 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 3 & 5 & 0 & 1 \end{array}\right] \downarrow$$

$$-3 \cdot R_1 + R_2 \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -1 & 3 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & -3 & -1 \end{array}\right] \downarrow$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -3 & -1 \end{array}\right] -2 \cdot R_2 + R_1$$

$x + 5z = 2$

$y - 3z = -1$

$x = 2 - 5z$

$y = -1 + 3z$

$\boxed{\infty \text{ Solutions}}$ $\xrightarrow{\text{parametrically}}$

$$\boxed{\begin{aligned} x &= 2-5t \\ y &= -1+3t \\ z &= t \end{aligned}}$$

Ex. 3: Use a system of equations to find the quadratic function $f(x) = ax^2 + bx + c$ that satisfies the equations using matrices.

$f(-1) = 8$, $f(1) = 2$, $f(3) = 4$

AP Calculus II
Notes P8.2
Matrix Multiplication

Another basic matrix operation is **matrix multiplication**. This operation becomes very applicable in retail and upper level math courses.

$m \times \underbrace{n \cdot n} \times p = m \times p$

Definition of Matrix Multiplication

If $A = [a_n]$ is an $m \times n$ matrix and $B = [b_n]$ is an $n \times p$ matrix, then the product AB is a $m \times p$ matrix.

The definition of matrix multiplication is a row by column multiplication in which the entries in the row of matrix B is multiplied to the entries in the column of matrix A and then adding all the results.

$3\times\underbrace{2 \cdot 3}\times 2$

Ex. 1: Find the product AB using $A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}$.

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} A\ -9 & D\ 1 \\ B\ -4 & E\ 6 \\ C\ -15 & F\ 10 \end{bmatrix}$$

$$\boxed{\begin{bmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{bmatrix} = AB}$$

A: $(-1)(-3) + (3)(-4)$
A: $3 + -12 = -9$
B: $(4)(-3) + (-2)(-4)$
B: $-12 + 8 = -4$
C: $(5)(-3) + (0)(-4) = -15$
D: $(-1)(2) + (3)(1) = 1$

Ex. 2: Find the product BA using $A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}$.

$2\times \cancel{2 \cdot 3} \times 2$

$$\begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} - & \\ & \end{bmatrix}$$

$(-3)(-1) + (2)(4) + (5)???$

$\boxed{\text{Can't multiply}}$

$3\times 1 \cdot 1\times 3$ ✓

$1\times 3 \cdot 3\times 1$

Ex. 3: Find the product AB using $A=\begin{bmatrix}7\\11\\-3\end{bmatrix}$ and $B=\begin{bmatrix}1 & -5 & 2\end{bmatrix}$ and BA.

$$AB = \begin{bmatrix}7\\11\\-3\end{bmatrix}\begin{bmatrix}1 & -5 & 2\end{bmatrix} = \begin{bmatrix}7 & -35 & 14\\11 & -55 & 22\\-3 & 15 & -6\end{bmatrix}$$

$$BA = \begin{bmatrix}1 & -5 & 2\end{bmatrix}\begin{bmatrix}7\\11\\-3\end{bmatrix} = \begin{bmatrix}-54\end{bmatrix}$$

$7 + -55 + -6$

Identity Matrix

A square matrix ($n\times n$) that consists of 1's on its main diagonal and 0's elsewhere is called the **identity matrix** and is denoted by I_n.

$$I_4 = \begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix}$$

$X + 0 = X$

$X \cdot 1 = X$

Ex. 4: Multiply A by I_3 given $A=\begin{bmatrix}3 & -2 & 1\\1 & 17 & -65\\-1 & 2 & 183\end{bmatrix}$.

$A \cdot I_3$

$$\begin{bmatrix}3 & -2 & 1\\1 & 17 & -65\\-1 & 2 & 183\end{bmatrix}\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} = \begin{bmatrix}3 & -2 & 1\\1 & 17 & -65\\-1 & 2 & 183\end{bmatrix}$$

$(3)(1)+(-2)(0)+(1)(0)$

$(1)(0)+(17)(1)+(-65)(0)$

$(-1)(0)+(2)(0)+(183)(1)$

$(-1)(1)+(2)(0)+(183)(0)$

$$\boxed{A \cdot I = A}$$

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In this procedure, the elimination process is continued until the augmented matrix is in the reduced row-echelon form.

Ex. 1: Solve the following system of equations.

a)
$$\begin{aligned} x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{aligned}$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & -1 & -1 & -1 \end{array}\right] \begin{array}{l} \\ 1 \cdot R_1 + R_2 \\ -2 \cdot R_1 + R_3 \end{array}$$

$$\downarrow$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & \boxed{1} & 3 & 5 \\ 0 & 0 & 2 & 4 \end{array}\right] \begin{array}{l} \\ \\ 1 \cdot R_2 + R_3 \end{array}$$

$$\leftarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

go back up

$$\begin{array}{l} -3 \cdot R_3 + R_1 \\ -3 \cdot R_3 + R_2 \\ \\ \end{array} \left[\begin{array}{ccc|c} 1 & -2 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right] \begin{array}{l} 2 \cdot R_2 + R_1 \\ \\ \\ \end{array}$$

$x = 1$
$y = -1$
$z = 2$

b)
$$\begin{aligned} a - b + 2c &= 4 \\ a + c &= 6 \\ 2a - 3b + 5c &= 4 \\ 3a + 2b - c &= 1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 1 & 0 & 1 & 6 \\ 2 & -3 & 5 & 4 \\ 3 & 2 & -1 & 1 \end{array}\right] \rightarrow \begin{array}{l} \\ -1 \cdot R_1 + R_2 \\ -2 \cdot R_1 + R_3 \\ -3 \cdot R_1 + R_4 \end{array} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & -1 & 1 & -4 \\ 0 & 5 & -7 & -11 \end{array}\right]$$

✓ 1 done

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & -2 \end{array}\right] \quad R_2 + R_3$$

$0 = -2$

never

$\therefore$ No Solution

Ex. 2: Solve the following system of equations:

$$2x + 4y - 2z = 0$$
$$3x + 5y = 1$$

$= \left[\begin{array}{ccc|c} 2 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{array}\right]$

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 3 & 5 & 0 & 1 \end{array}\right]$

$\downarrow$

$-3 \cdot R_1 + R_2 \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -1 & 3 & 1 \end{array}\right]$

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & -3 & -1 \end{array}\right]$

$\downarrow$

$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -3 & -1 \end{array}\right]$ $-2 \cdot R_2 + R_1$

$x + 5z = 2$

$y - 3z = -1$

$x = 2 - 5z$

$y = -1 + 3z$

$\boxed{\infty \text{ Solutions}} \xrightarrow{\text{parametrically}}$ $\boxed{\begin{array}{l} x = 2 - 5t \\ y = -1 + 3t \\ z = t \end{array}}$

Ex. 3: Use a system of equations to find the quadratic function $f(x) = ax^2 + bx + c$ that satisfies the equations using matrices.

$f(-1) = 8$, $f(1) = 2$, $f(3) = 4$

AP Calculus II
Notes P8.2
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Another basic matrix operation is **matrix multiplication**. This operation becomes very applicable in retail and upper level math courses.

$m \times n \cdot n \times p = m \times p$

Definition of Matrix Multiplication

If $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{ij}]$ is an $n \times p$ matrix, then the product AB is a $m \times p$ matrix.

The definition of matrix multiplication is a row by column multiplication in which the entries in the row of matrix B is multiplied to the entries in the column of matrix A and then adding all the results.

$3\times2 \cdot 2\times2$

Ex. 1: Find the product AB using $A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}$.

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} A: -9 & D: 1 \\ B: -4 & E: 6 \\ C: -15 & F: 10 \end{bmatrix}$$

$$\boxed{\begin{bmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{bmatrix} = AB}$$

A: $(-1)(-3) + (3)(-4)$
A: $3 + -12 = -9$
B: $(4)(-3) + (-2)(-4)$
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C: $(5)(-3) + (0)(-4) = -15$
D: $(-1)(2) + (3)(1) = 1$

Ex. 2: Find the product BA using $A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}$.

$\cancel{2\times2 \cdot 3\times2}$

$$\begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} - & \\ & \end{bmatrix}$$

$(-3)(-1) + (2)(4) + (5)???$

$\boxed{\text{Can't multiply}}$

$3 \times 1 \cdot 1 \times 3$ ✓

$1 \times 3 \cdot 3 \times 1$

Ex. 3: Find the product AB using $A = \begin{bmatrix} 7 \\ 11 \\ -3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -5 & 2 \end{bmatrix}$ and BA.

$$\begin{bmatrix} 1 & -5 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 7 \\ 11 \\ -3 \end{bmatrix} \begin{bmatrix} 7 & -35 & 14 \\ 11 & -55 & 22 \\ -3 & 15 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 7 \\ 11 \\ -3 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & -5 & 2 \end{bmatrix} \begin{bmatrix} -54 \end{bmatrix}$$

$7 + -55 + -6$

Identity Matrix

A square matrix ($n \times n$) that consists of 1's on its main diagonal and 0's elsewhere is called the **identity matrix** and is denoted by I_n.

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$X + 0 = X$

$X \cdot 1 = X$

Ex. 4: Multiply A by I_3 given $A = \begin{bmatrix} 3 & -2 & 1 \\ 1 & 17 & -65 \\ -1 & 2 & 183 \end{bmatrix}$.

$A \cdot I_3$

$(3)(1) + (-2)(0) + (1)(0)$

$(1)(0) + (17)(1) + (-65)(0)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -2 & 1 \\ 1 & 17 & -65 \\ -1 & 2 & 183 \end{bmatrix} \begin{bmatrix} 3 & -2 & 1 \\ 1 & 17 & -65 \\ -1 & 2 & 183 \end{bmatrix}$$

$(-1)(0) + (2)(0) + (183)(1)$

$(-1)(1) + (2)(0) + (183)(0)$

$$\boxed{A \cdot I = A}$$

$$\begin{aligned} x + 7 &= 8 \\ -7 &\quad -7 \\ x + 0 &= 1 \\ x &= 1 \end{aligned}$$

$AI = A$

AP Calculus II
Notes P8.3
Inverse of a Square Matrix

To start, remember how useful inverses are when solving equations. When inverse operations are applied to any equation, the result is an identity (adding 0 or multiplying by 1). The same is true for matrices.

Definition of the Inverse of a Square Matrix

A^{-1} is the inverse matrix of A if $AA^{-1} = I = A^{-1}A$ where I is the identity matrix.

Note that non-square matrices do not have an inverse and not all square matrices have an inverse. If a matrix does have an inverse, it is referred to as being **invertible** or **nonsingular**.

Ex. 1: Show that the following matrices are inverses: $A = \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}$.

$AB = BA = I$

AB

$$\begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

I_2

BA

$$\begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

I_2

$AB = BA = I, \quad A = B^{-1}, \quad B = A^{-1}$

Finding Inverse Matrices

$AA^{-1} = I$

Ex. 2: Find the inverse of the following matrices:

a) $A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}$

$$\left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ -1 & -3 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{array}\right] \; R_1+R_2$$

$$\left[\begin{array}{cc|cc} 1 & 0 & -3 & -4 \\ 0 & 1 & 1 & 1 \end{array}\right] \; -4R_2+R_1$$

$$\boxed{A^{-1} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}}$$

$$\begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$a+4c=1 \qquad b+4d=0$

$-a-3c=0 \;, \; -1b-3d=1$

$$\left[\begin{array}{cc|c} 1 & 4 & 1 \\ -1 & -3 & 0 \end{array}\right] , \left[\begin{array}{cc|c} 1 & 4 & 0 \\ -1 & -3 & 1 \end{array}\right]$$

b) $B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 5 & 6 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{-5R_1+R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & -4 & -15 & -5 & 0 & 1 \end{array}\right]$$

$4R_2+R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array}\right]$$

$-3R_3+R_1$

$-4R_3+R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 16 & -12 & -3 \\ 0 & 1 & 0 & 20 & -15 & -4 \\ 0 & 0 & 1 & 5 & 4 & 1 \end{array}\right]$$

$-2R_2+R_1$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \boxed{\begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ 5 & 4 & 1 \end{bmatrix}}$$

$$\begin{array}{r} 18 \\ -30 \\ \hline -12 \end{array} \qquad \begin{array}{r} 25 \\ -24 \\ \hline 1 \end{array}$$

$$\begin{array}{r} -120 \\ 120 \\ \hline 0 \end{array}$$

$$\boxed{\begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ 5 & 4 & 1 \end{bmatrix}}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

c) $M = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}$

$\boxed{M^{-1} = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}}$

$15/2 + \frac{-14}{2}$

$-15/2 + 1/2$

$\left[\begin{array}{cc|cc} 2 & 5 & 1 & 0 \\ -3 & -7 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 5/2 & 1/2 & 0 \\ -3 & -7 & 0 & 1 \end{array}\right]$ $3R_1 + R_2$

$\rightarrow \left[\begin{array}{cc|cc} 1 & 5/2 & 1/2 & 0 \\ 0 & 1/2 & 3/2 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 5/2 & 1/2 & 0 \\ 0 & 1 & 3 & 2 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & -7 & -5 \\ 0 & 1 & 3 & 2 \end{array}\right]$

$-5/2 R_2 + R_1$

d) $C = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ -2 & 3 & -2 \end{bmatrix}$

$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ -2 & 3 & -2 & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & -7 & 2 & -3 & 1 & 0 \\ 0 & 7 & -2 & 2 & 0 & 1 \end{array}\right]$ $\begin{array}{l} \\ -3R_1 + R_2 \\ 2R_1 + R_3 \end{array}$

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & -2/7 & 3/7 & -1/7 & 0 \\ 0 & 7 & -2 & 2 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & -2/7 & 3/7 & -1/7 & 0 \\ 0 & 0 & \boxed{0} & -1 & 1 & 1 \end{array}\right]$

$-7R_2 + R_3$

Uh-oh . . . . . .

~~ook?~~

NO inverse

$\boxed{\text{noninvertible}}$

AP Calculus II
Notes P8.4
Determinant of a Square Matrix

Every square matrix can be associated with a feature known as the **determinant**. This feature has a lot of special uses in Linear Algebra and this use arose from special number patterns that occur when system of equations are solved.

$-\frac{bc}{a} + \frac{ad}{a}$

Proof of the Determinant of a 2x2 Matrix

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ c & d & 0 & 1 \end{array}\right] \xrightarrow{-c\cdot R_1 + R_2} \left[\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & \frac{ad-bc}{a} & -\frac{c}{a} & 1 \end{array}\right]$$

$$\left[\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right] \xleftarrow{R_2 \cdot \frac{a}{ad-bc}}$$

$$\xrightarrow{-\frac{b}{a}\cdot R_2 + R_1} \left[\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right]$$

$$\frac{\cancel{bc}}{a(ad-bc)} + \frac{ad-\cancel{bc}}{a(ad-bc)}$$

$$\frac{\cancel{a}d}{\cancel{a}(ad-bc)}$$

for any $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \to \boxed{M^{-1} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}}$

$\boxed{ad-bc}$

Definition of the Determinant of a 2x2 Matrix

The determinant of the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is given by $\det(A) = |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} =$ $ad - bc$

Ex. 1: Find the determinant of the following matrices:

a) $A=\begin{bmatrix}2 & -3\\ 4 & 1\end{bmatrix}$

$|A| = \begin{vmatrix}2 & -3\\ 4 & 1\end{vmatrix}$

$= (2)(1)-(-3)(4)$

$= \boxed{14}$

b) $B=\begin{bmatrix}1 & -2\\ -3 & 6\end{bmatrix}$

$\det(B) = 6-6$

$= 0$

B is singular, $\boxed{B^{-1} \text{ does not exist}}$

c) $C=\begin{bmatrix}0 & 2\\ 3 & 8\end{bmatrix}$.

$|C| = -6$

Determinant of a 3x3 Matrix

Unfortunately for us, the proof of this is a little too much fun for one day, so here it is:

$$\det(A)=\begin{vmatrix}a & b & c\\ d & e & f\\ g & h & i\end{vmatrix}=a(ei-fh)-b(di-fg)+c(dh-eg)$$

$$a\begin{vmatrix}e & f\\ h & i\end{vmatrix}-b\begin{vmatrix}d & f\\ g & i\end{vmatrix}+c\begin{vmatrix}d & e\\ g & h\end{vmatrix}$$

A common visual approach to the determinant of a 3x3 matrix is:

$$\begin{matrix}a & b & c & a & b\\ d & e & f & d & e\\ g & h & i & g & h\end{matrix} = aei+bfg+cdh-afh-bdi-ceg$$

$$a(ei-fh)-b(di-fg)+c(dh-eg)$$

Ex. 2: Find the determinant of the following matrices:

a) $\begin{bmatrix} 5 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix} \rightarrow 5\begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} - 2\begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} + 1\begin{vmatrix} 3 & -1 \\ 4 & 0 \end{vmatrix}$

$$5(-1-0)-2(3-8)+1(0-{}^-4)$$

$$-5+10+4=\boxed{9}$$

$$\begin{matrix} 5 & 2 & 1 & 5 & 2 \\ 3 & -1 & 2 & 3 & -1 \\ 4 & 0 & 1 & 4 & 0 \end{matrix} = -5+16+0-6-0-{}^-4=\boxed{9}$$

b) $\begin{bmatrix} 0 & 3 & 8 \\ 3 & -2 & 2 \\ -7 & 5 & 0 \end{bmatrix}$

$$0\begin{vmatrix} -2 & 2 \\ 5 & 0 \end{vmatrix} - 3\begin{vmatrix} 3 & 2 \\ -7 & 0 \end{vmatrix} + 8\begin{vmatrix} 3 & -2 \\ -7 & 5 \end{vmatrix}$$

$$0-3(0-{}^-14)+8(15-14)$$

$$=-42+8=\boxed{-34}$$

Cryptography

Encryption matrix - $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix} = E$ Decryption matrix - $\begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$

$M \to \boxed{M \cdot E}\ D \to M(E \cdot E^{-1}) \to M \cdot I \to M$

Take a 9-character message and put it in a 3x3 matrix.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

27 28 29 30 ... 53
54 ... 80
81 ... 107
108

$3 \times 3 \quad 3 \times 3$

$$M \cdot E = \begin{bmatrix} 23 & 5 & 12 \\ 12 & 0 & 4 \\ 15 & 14 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix} = \begin{bmatrix} 83 & 123 & 89 \\ 32 & 48 & 36 \\ 40 & 74 & 101 \end{bmatrix}$$

$$\begin{bmatrix} B & O & H \\ E & U & I \\ M & T & T \end{bmatrix}$$

-1992
2460
468
-445

$\cancel{144}$ $\cancel{576}$ 720
-720

$$\begin{bmatrix} 83 & 123 & 89 \\ 32 & 48 & 36 \\ 40 & 74 & 101 \end{bmatrix} \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 23 & 5 & 12 \\ 12 & 0 & 4 \\ 15 & 14 & 5 \end{bmatrix} \to \begin{bmatrix} W & E & L \\ L & W & D \\ O & N & E \end{bmatrix}$$

Calculus Linear Algebra Practice Test

Name: ____________________________

For the following questions, a calculator may be used.

1. Solve the following system of equations. If there are no solutions, state so and if there are infinite solutions, be sure to describe the relationship of the solutions.

a)
$$\begin{aligned} x \qquad - 3z &= -2 \\ 3x + y - 2z &= 5 \\ 2x + 2y + z &= 4 \end{aligned}$$

b)
$$\begin{aligned} x + 2y &= 7 \\ 2x + y &= 8 \end{aligned}$$

c)
$$\begin{aligned} x + y + 4z &= 5 \\ 2x + y - z &= 9 \end{aligned}$$

d)
$$\begin{aligned} 2x + y - z &= 2 \\ x + y - 3z &= 4 \\ 3x + 2y - 4z &= 7 \end{aligned}$$

2. Find AB and then BA, if possible.

a) $A = \begin{bmatrix} 2 & 1 \\ 4 & -3 \\ 1 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 5 \\ 3 & -2 & 0 \end{bmatrix}$

b) $A = \begin{bmatrix} 1 & 7 & -2 \\ 0 & 4 & 3 \end{bmatrix}, B = \begin{bmatrix} -3 \\ 4 \\ 5 \end{bmatrix}$

3. Are the following matrices inverses?

$$N = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix} \quad , \quad O = \begin{bmatrix} 2 & -17 & 11 \\ -1 & 11 & -7 \\ 0 & 3 & -2 \end{bmatrix}$$

4. Find the inverse of:

a) $\begin{bmatrix} 1 & -2 \\ 2 & -3 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 4 \\ 3 & 6 & 5 \end{bmatrix}$

5. Find the determinant of the following:

a) $\begin{bmatrix} 7 & -2 \\ \frac{1}{2} & -3 \end{bmatrix}$

b) $\begin{bmatrix} 3\cos\theta & -2\sin\theta \\ 6\sin\theta & 4\cos\theta \end{bmatrix}$

c) $\begin{bmatrix} 1 & 2 & 1 \\ 3 & 5 & -4 \\ 1 & 6 & 5 \end{bmatrix}$

Solutions:

1. a) $$\begin{aligned} x \quad\quad - 3z &= -2 \\ 3x + y - 2z &= 5 \\ 2x + 2y + z &= 4 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 3 & 1 & -2 & 5 \\ 2 & 2 & 1 & 4 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 0 & 1 & 7 & 11 \\ 0 & 2 & 7 & 8 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 0 & 1 & 7 & 11 \\ 0 & 0 & -7 & -14 \end{array}\right] \to$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 0 & 1 & 7 & 11 \\ 0 & 0 & 1 & 2 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$x = 4\,, y = -3\,, z = 2$

b) $$\begin{aligned} x + 2y &= 7 \\ 2x + y &= 8 \end{aligned}$$

$$\left[\begin{array}{cc|c} 1 & 2 & 7 \\ 2 & 1 & 8 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & -3 & -6 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & 2 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \end{array}\right]$$

$x = 3\,, y = 2$

c) $$\begin{aligned} x + y + 4z &= 5 \\ 2x + y - z &= 9 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 4 & 5 \\ 2 & 1 & -1 & 9 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 4 & 5 \\ 0 & -1 & -9 & -1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 4 & 5 \\ 0 & 1 & 9 & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & -5 & 4 \\ 0 & 1 & 9 & 1 \end{array}\right]$$

Which means $\begin{aligned} x - 5z &= 4 \\ y + 9z &= 1 \end{aligned}$, so there are infinite solutions. This can be expressed parametrically as $x = 4 + 5t\,, y = 1 - 9t\,, z = t$

d) $$\begin{aligned} 2x + y - z &= 2 \\ x + y - 3z &= 4 \\ 3x + 2y - 4z &= 7 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & 2 \\ 2 & 1 & -1 & 4 \\ 3 & 2 & -4 & 7 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & -3 & 2 \\ 0 & -1 & 5 & 0 \\ 0 & -1 & 5 & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & -3 & -2 \\ 0 & 1 & -5 & 0 \\ 0 & -1 & 5 & 1 \end{array}\right] \to$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & -2 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$ The last line is a contradiction, so no solution.

2.

a)

$$AB = \begin{array}{cc} & \begin{bmatrix}1&0&5\\3&-2&0\end{bmatrix} \\ \begin{bmatrix}2&1\\4&-3\\1&6\end{bmatrix} & \begin{bmatrix}5&-2&10\\-5&6&20\\19&-12&5\end{bmatrix}\end{array}$$

$(1)(2)+(3)(1)=5$

$(4)(0)+(-3)(-2)=6$

$(1)(5)+(6)(0)=5$

$$BA = \begin{array}{cc} & \begin{bmatrix}2&1\\4&-3\\1&6\end{bmatrix} \\ \begin{bmatrix}1&0&5\\3&-2&0\end{bmatrix} & \begin{bmatrix}7&31\\-2&9\end{bmatrix}\end{array}$$

$(1)(2)+(0)(4)+(5)(1)=7$

$(3)(1)+(-2)(-3)+(0)(6)=9$

b)

$$AB = \begin{array}{cc} & \begin{bmatrix}-3\\4\\5\end{bmatrix} \\ \begin{bmatrix}1&7&-2\\0&4&3\end{bmatrix} & \begin{bmatrix}15\\31\end{bmatrix}\end{array}$$

$(1)(-3)+(7)(4)+(-2)(5)=15$

$(0)(-3)+(4)(4)+(3)(5)=31$

$$BA = \begin{array}{cc} & \begin{bmatrix}1&7&-2\\0&4&3\end{bmatrix} \\ \begin{bmatrix}-3\\4\\5\end{bmatrix} & \begin{bmatrix}?&?&?\\?&?&?\\?&?&?\end{bmatrix}\end{array}$$

One number in the row and two in the column, can't multiply.

3. Matrices are inverses if $NO = I_3$

$$NO = \begin{bmatrix}1 & 1 & 2\\ 2 & 4 & -3\\ 3 & 6 & -5\end{bmatrix}\begin{bmatrix}2 & -17 & 11\\ -1 & 11 & -7\\ 0 & 3 & -2\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

So yes they are inverses

4.

a) $\begin{bmatrix}1 & -2\\ 2 & -3\end{bmatrix}$ $\left[\begin{array}{cc|cc}1 & -2 & 1 & 0\\ 2 & -3 & 0 & 1\end{array}\right] \rightarrow \left[\begin{array}{cc|cc}1 & -2 & 1 & 0\\ 0 & 1 & -2 & 1\end{array}\right] \rightarrow \left[\begin{array}{cc|cc}1 & 0 & -3 & 2\\ 0 & 1 & -2 & 1\end{array}\right]$ $\begin{bmatrix}-3 & 2\\ -2 & 1\end{bmatrix}$

b) $$\left[\begin{array}{ccc|ccc}1 & 1 & 1 & 1 & 0 & 0\\ 3 & 5 & 4 & 0 & 1 & 0\\ 3 & 6 & 5 & 0 & 0 & 1\end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc}1 & 1 & 1 & 1 & 0 & 0\\ 0 & 2 & 1 & -3 & 1 & 0\\ 0 & 3 & 2 & -3 & 0 & 1\end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc}1 & 1 & 1 & 1 & 0 & 0\\ 0 & 1 & \frac{1}{2} & \frac{-3}{2} & \frac{1}{2} & 0\\ 0 & 3 & 2 & -3 & 0 & 1\end{array}\right] \rightarrow$$

$$\left[\begin{array}{ccc|ccc}1 & 1 & 1 & 1 & 0 & 0\\ 0 & 1 & \frac{1}{2} & \frac{-3}{2} & \frac{1}{2} & 0\\ 0 & 0 & \frac{1}{2} & \frac{3}{2} & \frac{-3}{2} & 1\end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc}1 & 1 & 1 & 1 & 0 & 0\\ 0 & 1 & \frac{1}{2} & \frac{-3}{2} & \frac{1}{2} & 0\\ 0 & 0 & 1 & 3 & -3 & 2\end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc}1 & 1 & 0 & -2 & 3 & -2\\ 0 & 1 & 0 & -3 & 2 & -1\\ 0 & 0 & 1 & 3 & -3 & 2\end{array}\right] \rightarrow$$

$$\left[\begin{array}{ccc|ccc}1 & 1 & 0 & 1 & 1 & -1\\ 0 & 1 & 0 & -3 & 2 & -1\\ 0 & 0 & 1 & 3 & -3 & 2\end{array}\right] \qquad \begin{bmatrix}1 & 1 & -1\\ -3 & 2 & -1\\ 3 & -3 & 2\end{bmatrix}$$

5.

a) $d = (7)(-3) - (-2)\left(\frac{1}{2}\right) = -20$

b) $d = (3\cos\theta)(4\cos\theta) - (-2\sin\theta)(6\sin\theta) = 12\cos^2\theta + 12\sin^2\theta = 12$

c) $d = 1\begin{vmatrix}5 & -4\\ 6 & 5\end{vmatrix} - 2\begin{vmatrix}3 & -4\\ 1 & 5\end{vmatrix} + 1\begin{vmatrix}3 & 5\\ 1 & 6\end{vmatrix} = 1(25+24) - 2(15+4) + 1(18-5) = 24$